Applied A.I. Solutions
Foundations of Data Management

Exercises + Project objectives: to generate an Operational Report and an Executive Report under a Data Governance Framework in which exercises 1, 2, and 3 are the building blocks towards the Project development.

Lab Exercise 1 - (10 points max)

Groups will use **Sample – Superstore.xls** to generate and complete the exercise deliverables (reports). Sample – Superstore.xls is available on D2L/Assignments.

1. **Operational Report** and the **Executive Report –** explain:

- Business Analysis: Context + Framework

  a. read and understand the spreadsheet's data
  b. on a document, indicate sector/industry of study, identify the entities with impact on the reporting process
  c. define the target audience for the reports
  d. indicate the intended use of each report (monitor and control, decision-making, performance improvement, research & analysis, etc.)
  e. indicate the publishing and distribution frequency, maintenance, and use of the reports
  f. indicate any assumptions you need, in order to produce the correct deliverables in a timely manner
  g. indicate the information that both Operational Report and Executive Report will include (i.e.: sales, geography, customers, revenue, KPIs, etc.)

- Technical Analysis:

  h. identify and document completeness, inconsistency, redundancies, duplicates
  i. indicate any additional calculations (formulas/algorithms), KPIs (ratios), new entities, other attributes etc., to incorporate to the spreadsheet
  j. clean/transform the data source for further use and data analysis. Document every change (field update, deletion, override, etc.) performed on the spreadsheet.

  Note: groups can use SQL and/or Python code

2. **Report form designs (w/o data)**: design the Operational Report form and the Executive Report form, indicating titles, column titles, date/period, units, author, versioning. Include in the design the columns for comparisons and columns with KPIs. For report examples, refer to **Annex I and II** below

3. **Submission**: document produced in 1, 2 - Submit **one pdf** document as **Group#_1.pdf**

Lab Exercise 2 - (10 points max)

1. **Advanced Data Analysis**:

- Identify, document entities, attributes, domains, referential integrity (primary, foreign keys)

- Data/Process Flow Diagram (DFD): create a data / process flow diagram indicating data in/out to/from processes, sequence and relationship among processes, process ownership, and data/ process orchestration. Note: use Lucidchart online

- Logical-level ERD: create the corresponding ERD indicating entities, attributes, relationships, primary and foreign keys, and Master Data tables. Include a Metadata table and a Reference Data table. Use crows' foot notation and take the model to 2NF. Note: use Lucidchart online

- Model Alignment: transform Sample–Superstore.xls to align the physical table to the previously created logical ERD. To get the alignment right, you may need to split Superstore into several new tables, entities and attributes. Document every change (field change, update, deletion, override, etc.) performed on the spreadsheet. **Note**: groups can use SQL and/or Python code

2. **Submission:**

   a. **DFD**: Data / Process Flow Diagram from Lucidchart (export as PDF, <u>no links</u>)

   b. **ERD**: Logical-level ERD from Lucidchart (export as PDF, <u>no links</u>)

   Submit **one pdf** document as **Group#_2.pdf**

Lab Exercise 3 - (10 points max) - Note: to answer the exercise, groups can use **SQL** and/or **Python** (*)

1. Database installation: install MySQL Workbench in your computer

2. Database creation: using your ERD/2NF as foundation, create an aligned physical MySQL database schema named "GBC_Superstore".

3. Extract, Transform, Load (ETL): data from GBC_Superstore.xls to MySQL database GBC_Superstore You may follow the following steps:

    a. From GBC_Superstore.xlsx, create & save separate .csv files per each table (tabs) you got from the model designed during exercises 2

    b. Then, import each .csv file onto MySQL database/tables using Table Data Import Wizard option (one file at a time)

4. Using MySQL database GBC_Superstore quality data, based on your previous report form designs, produce both Operational Report and Executive Report. You may follow the following steps:

    a. once MySQL GBC_Superstore tables have been populated, create a temp table called "operational_report" and a temp table called "executive_report" with the columns (attributes) needed to generate your reports; minor changes can still be made later; read d) below. Note: temp tables cannot be generated using excel

    b. use SQL scripts and/or Python code to fill in and complete each temp table

    c. export "operational_report" and "executive_report" as .csv files

    d. using cleaned data only from GBC_Superstore database, use Excel, and the forms designed in Ex 1, to open "operational_report.csv" and "executive_report.csv" files and produce the reports

5. **Submission**:

    a. Operational Report and Executive Report as one PDF named Group#_3R.pdf
    b. ETL process, step-by-step detailed explanation (use numbered bullets), including SQL scripts and/or Python code, as one PDF named Group#_3ETL.pdf
    c. Temp "Operational_Report.csv" and "Executive_Report.csv" used to generate the reports

(*) Note: students who do not have SQL experience can read **SQL basics** and self-study using the content on D2L. Students who know SQL may share with and transfer their knowledge to the classmates.

PROJECT - (30 points max)

1. **Apply a Data Management Framework to the Operational and Executive Reporting process**

   Data Governance Framework: based on the business analysis, data modeling and execution of exercises 1, 2 and 3, groups must apply theory into practice to define the **principles, policies and procedures,** in alignment to the DAMA Wheel, needed to govern the reporting process

   Elements to consider:

   - Business Assumptions: formulate business assumptions about the business, stakeholders, audience, technology, reporting, processes, report objectives, etc.

   - Quality Data: elaborate on the how your reports will present the right information to a target audience, with the appropriate context to facilitate a correct interpretation and understanding of its quality data, effectively, securely, and in a timely manner

   using your own words … (do not include theory, do not use AI apps)

   a. identify 2 (two) core sub-domains from the DAMA Wheel. Explain your selection

   b. for each sub-domain, write 2 (two) principles to enforce oversight of the reporting. The objective is to achieve a well-controlled, transparent, efficient, quality-driven process.

   c. for each principle, write 2 (two) policies to enforce oversight of the reporting. The objective is to achieve a well-controlled, transparent, efficient, quality-driven process.

   d. for each policy, write 1 (one) procedure that you would implement, and under which you will generate the operational and executive reports on a regular basis.

   2 Sub-domains → 2 Principles x sub-domain → 1 Policy x Principle → 1 Procedure x Policy

2. **Submission**: use one PDF file to produce a document including:

   a. Cover Page (Indicate group # and team members who contributed to the project)
   b. Document including points a. + b. + c. + d.
   c. Submit as **Group#_P.pdf**

- **MySQL installation**
  - Install MySQL V 8.X.X Win64 (x86_64) (MYSQL Community Server – GPL)
  - Install MySQL Workbench 8.X - V 8.X Community

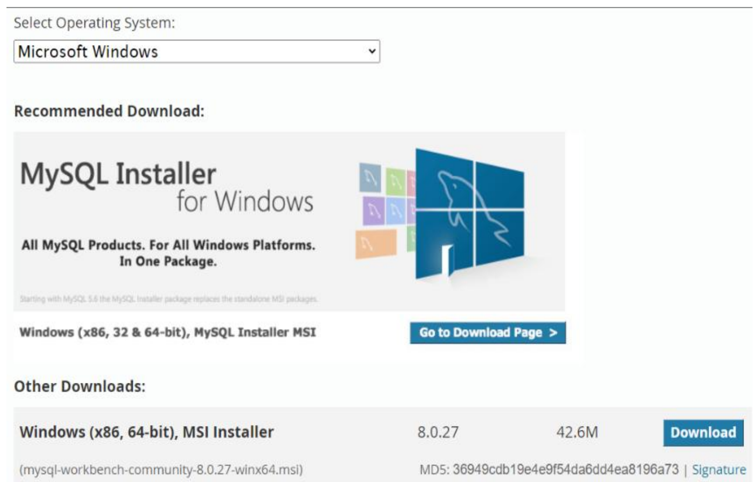
  https://dev.mysql.com/downloads/workbench/

  or

  https://dev.mysql.com/downloads/

  MySQL parameters

  - Server: localhost
  - Username: root
  - Password: 1234
  - Port 3306



Important

At the end of this project, **DO NOT** uninstall the software from your computers. Write down installation parameters, user IDs, passwords for later use. We will use these tools and deliverables again during the Data Visualization Techniques course.

EXAMPLE

ANNEX I

Sales Team Operations by Quarter

01/01/22 - 03/31/22

Region	Province	City	Sales per person	Item	Unit Price ($)	Quantity Sold	Quantity Quota	Quantity Sold / Quantity Quota (%)	Discount Applied (%)	Discount Cap (%)	Gross Sales ($)	Gross Sales Quota ($)	Gross Sales / Gross Sales Quota (%)
	New Brunswick	Moncton	S1	Chair	350	50.00	45.00	11%	10%	10%	$15,750	$15,750	0%
				Desk	750	28.00	30.00	-7%	-5%	5%	$22,050	$22,500	-2%
				Bookshelf	1,200	32.00	30.00	7%	5%	7%	$36,480	$36,000	1%
		Subtotal									$74,280	$74,250	0%
	Newfoundland and Labrador	St. John's	S2	Chair	350	30.00	30.00	0%	10%	10%	$9,450	$10,500	-10%
				Desk	750	18.00	20.00	-10%	-3%	5%	$13,905	$15,000	-7%
				Bookshelf	1,200	10.00	16.00	-38%	5%	7%	$11,400	$19,200	-41%
		Subtotal									$34,755	$44,700	-22%
Atlantic	Nova Scotia	Halifax	S3	Chair	350	50.00	45.00	11%	10%	10%	$15,750	$15,750	0%
				Desk	750	35.00	30.00	17%	-7%	10%	$28,088	$22,500	25%
				Bookshelf	1,200	32.00	30.00	7%	5%	10%	$36,480	$36,000	1%
		Subtotal									$80,318	$74,250	8%
	Prince Edward Island	Charlottetown	S4	Chair	350	40.00	30.00	33%	10%	10%	$12,600	$10,500	20%
				Desk	750	20.00	25.00	-20%	0%	4%	$15,000	$18,750	-20%
				Bookshelf	1,200	10.00	10.00	0%	5%	9%	$11,400	$12,000	-5%
		Subtotal									$39,000	$41,250	-5%
	Total										$228,353	$234,450	-3%
	Ontario	Toronto	S5	Chair	350	50.00	45.00	11%	10%	10%	$15,750	$15,750	0%
				Desk	750	28.00	30.00	-7%	-5%	5%	$22,050	$22,500	-2%
				Bookshelf	1,200	32.00	30.00	7%	5%	7%	$36,480	$36,000	1%
		Subtotal									$74,280	$74,250	0%
		Ottawa	S6	Chair	350	30.00	30.00	0%	10%	10%	$9,450	$10,500	-10%
				Desk	750	18.00	20.00	-10%	-3%	5%	$13,905	$15,000	-7%
				Bookshelf	1,200	10.00	16.00	-38%	5%	7%	$11,400	$19,200	-41%
		Subtotal									$34,755	$44,700	-22%
		Total									$109,035	$118,950	-8%

Central	Quebec	Montreal	S7	Chair	350	50.00	45.00	11%	10%	10%	$15,750	$15,750	0%
				Desk	750	35.00	30.00	17%	-7%	10%	$28,088	$22,500	25%
				Bookshelf	1,200	32.00	30.00	7%	5%	10%	$36,480	$36,000	1%
		Subtotal									$80,318	$74,250	8%
		Quebec City	S8	Chair	350	40.00	30.00	33%	10%	10%	$12,600	$10,500	20%
				Desk	750	20.00	25.00	-20%	0%	4%	$15,000	$18,750	-20%
				Bookshelf	1,200	10.00	10.00	0%	5%	9%	$11,400	$12,000	-5%
		Subtotal									$39,000	$41,250	-5%
		Total									$119,318	$115,500	3%
	Total										$228,353	$234,450	-3%
TOTAL											$456,705	$468,900	-3%

Notes:

1. Breakdown by region, province, city, salespersons, items
2. Note all subtotals and grans totals
3. The operational report can be transformed based on the stakeholders' needs.
4. Breakdown is possible by items / salespeople / region / etc.
5. It is only an example of the details that an Operational report should have
6. For distribution best practice use PDF formatting, letter size to facilitate printing it.
7. It is a best practice to assess users' needs to incorporate additional details along with comparisons and KPIs that can facilitate business process monitoring and controlling.

EXAMPLE

ANNEX II

Sales Team Monthly Executive Report
Gross Sales Revenue ($)

Province	Jan-22	Feb-22	Feb 22 vs Jan 22 (%)	Feb-21	Feb 22 vs Feb 21 (%)	KPIs: Participation Ratio AVG 2021	KPIs: Particpation Ratio Feb 22 (%)
New Brunswick	$51,250	$56,000	9.3%	$55,500	0.9%	30.0%	32.6%
New Foundland Labrador	$32,560	$35,000	7.5%	$34,958	0.1%	25.0%	20.4%
Nova Scotia	$65,900	$68,456	3.9%	$55,650	23.0%	35.0%	39.8%
Prince Edward Island	$15,984	$12,365	-22.6%	$10,500	17.8%	10.0%	7.2%
East Region	$165,694	$171,821	3.7%	$156,608	9.7%	100.0%	100.0%

Notes:

1. mainly aggregated data to facilitate decison-making
2. It is only an example of aggregation that any Executive Report should have
3. For distribution best practice use PDF formatting, letter size to facilitate printing it.
4. It is a best practice to assess users' needs to incorporate additional details along with comparisons and KPIs that can facilitate business process monitoring and controlling